Luiz Carlos Costa da Silva

AI Engineer · Senior Specialist | LLM Systems, AI Agents & RAG | Python, TypeScript, FastAPI

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Portuguese (Native) • English (Advanced) • Mandarin (Learning)

Summary

AI Engineer (Senior Specialist) building production LLM systems, AI agents, and RAG pipelines, plus the backend services and business process automation around them. Currently responsible for AI and automation initiatives at BWG. Previously built AI agent pipelines at Auria AI that process **3,000+ security incidents per day**.

Skills

Backend: Python, FastAPI, asyncio, Pydantic, SQLAlchemy, Node.js, Express, REST APIs, Microservices, Pytest
AI / LLMs: AI Agents, RAG, Embeddings, Prompt Engineering, Structured Outputs, LangChain, Pydantic-AI, Gemini, OpenAI, Claude
Data & Tools: PostgreSQL, SQLite, Redis, Docker, Kubernetes, Git, Metabase, Twilio API, Vector Search, CI/CD
Frontend: TypeScript, JavaScript, Vue.js, React, HTML, CSS

Experience

AI Engineer · Senior Specialist (Applied AI & Automation) August 2026 – Present
BWG – HR Technology & Corporate Benefits São Paulo, Brazil

- Own AI and automation initiatives for the corporate health-insurance brokerage, from process discovery to deployment and adoption, reporting directly to the CEO.
- Mapped the quoting and client-onboarding workflow step by step, classifying each step as API integration, deterministic automation, LLM assistance, or human review. Defined baseline metrics to evaluate the MVP.
- Built an internal platform (TypeScript, React, Node.js, PostgreSQL) that combines hospital-network coverage analysis, pricing automation on **four insurers**' portals, quote-request forms for **five insurers**, and onboarding document generation. After the initial demo, the team asked to adopt it as their official channel for quote requests.
- Built a Gemini-based extraction pipeline for client PDFs, spreadsheets, and scanned documents. Each extracted field links back to its source, missing data is flagged instead of guessed, and pricing calculations stay outside the LLM.
- Deployed and operate the platform on Kubernetes, with CI/CD pipelines and automated backups.

Backend Software Engineer August 2025 – July 2026
Auria AI São Paulo, Brazil

- Built and maintained **20+ Python AI agents** in the production incident-response pipeline, handling summarization, insight extraction, and automated decision-making for **3,000+ B2B security incidents per day**.
- Bootstrapped and maintained the PostgreSQL/SQLAlchemy data layer across **36 production tables**, shipping backend features end to end across multiple FastAPI services.
- Built **Auria's Metabase BI from scratch**, including operational dashboards, KPI analyses, and threshold-based alerts used daily by incident-response leadership.
- Designed and shipped an **on-call alerting bot** that calls the responsible engineers through Twilio with AI-generated incident summaries, reducing overnight outage detection time from **8+ hours to near-immediate**.

Software Developer (Research Scholarship) January 2021 – August 2023
Hub USP Inovação – IME-USP São Paulo, Brazil

- Maintained the innovation platform of AUSPIN (USP's Innovation Agency), which catalogs **1,000+ patents, 3,000+ companies, 3,000+ professors, and 100+ innovation initiatives**.
- Migrated the platform's backend from a legacy Ruby on Rails system to Kotlin, redesigning domain models, REST APIs, automated tests, and Docker deployment infrastructure.

Projects

Memoir – AI Meeting Knowledge Workspace 2026 – Present

- Built an open-source full-stack application (FastAPI, Pydantic-AI, SQLAlchemy, JavaScript) that turns meeting transcripts into searchable summaries, decisions, action items, participant profiles, and structured transcript segments.
- Implemented automatic format detection and parsers for Zoom and Google Meet, with a Gemini fallback for messy or unsupported formats. Added Pytest coverage for parsing and core workflows.

Divination – RAG Assistant for D&D Dungeon Masters 2024

- B.Sc. capstone at IME-USP: a RAG assistant grounded in the official 2024 D&D Free Rules, built with LangChain, vector search, GPT-4o, FastAPI, and Vue 3.

Education

B.Sc. in Computer Science, University of São Paulo (IME-USP) 2020 – 2024

Leadership & Activities

President & Member, USPCodeLab January 2021 – March 2024
IME-USP São Paulo, Brazil

- Served as President from January 2022 to August 2023, leading **20+ members**; organized two university hackathons, secured sponsorships, mentored participants, and managed event budgets.